CHEMICAL EQUILIBRIUM

1. Equilibrium can be established –
(1) Only for physical processes
(2) Only for chemical processes
(3) for both physical & chemical processes
(4) None of these

2. At equilibrium –
(1) concentration of reactants and products are equal.
(2) concentration of reactants is double to that of products.
(3) concentration of reactants and products becomes constant.
(4) None of these

3. In the reaction $X(g) \rightleftharpoons 2Y(g)$, the concentration of X is increased, when –
(1) pressure is increased
(2) pressure is decreased
(3) inert gas is added at constant volume
(4) None of these

4. Which of the following is correct?
(1) At equilibrium rate of forward reaction is greater than that of rate of backward reaction.
(2) Equilibrium can be attained from either of the direction.
(3) On addition of catalyst the equilibrium state gets affected.
(4) None of these

5. Which of the following is correct for the reaction in which volume change has no effect –
(1) $K_p < K_c$ (2) $K_p = K_c$
(3) $K_p > K_c$ (4) None of these

6. For which of the following reaction K_p is unit less ?
(1) $CaCO_3(s) \rightleftharpoons CO_2(g) + CaO(s)$
(2) $Ni(s) + 4CO(g) \rightleftharpoons [Ni(CO)_4](g)$
(3) $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$
(4) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

7. If a reversible reaction shifting from right to left then which of the following is correct?
(1) $Q_p < K_p$ (2) $Q_p = K_p$
(3) $Q_p > K_p$ (4) None

8. What will happen if temperature is increased in the dissolution of CO_2 gas in water in a closed container?
(1) solubility decreases
(2) solubility increases
(3) solubility remains unchanged
(4) None of these

9. For the equilibrium $H_2O(\ell) \rightleftharpoons H_2O(g)$ attained in a closed container at 40°c. The vapour pressure of water is 23 mm of Hg. What is K_p for reaction?
(1) 46 mm of Hg (2) 23 mm of Hg
(3) 15 mm of Hg (4) data insufficient

10. The K_p for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, is 16. If pressure at equilibrium is doubled, then K_p will be –
(1) 32 (2) 8
(3) 24 (4) 16

11. In the reaction $A \rightleftharpoons B + C$, initially 3 moles of A were taken. At equilibrium 0.75 moles of B are formed. The degree of dissociation of A is –
(1) 0.75 (2) 0.25
(3) 0.50 (4) 0.40

12. The reactant will be most stable if the value of K_C for the reaction is –
(1) 2×10^{-7} (2) 5×10^{8}
(3) 1.7×10^{-3} (4) 1×10^{-15}

13. The K_c for the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ is 0.04. At a given time 0.1 M PCl_5, 0.3 M each of PCl_3 and Cl_2 are present then predict direction of reaction –
(1) forward (2) backward
(3) at equilibrium (4) None of these

14. If $K_c < 10^{-3}$ then
(1) The reaction proceeds nearly to completion
(2) The extent of reaction is very low
(3) Appreciable concentrations of both reactants & products are present
(4) None of these

15. For reaction

$H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ at 200 K if

K_c is 4.0×10^{31} then we conclude that–

(1) The reaction proceeds nearly to completion

(2) The reaction proceeds rarely

(3) Appreciable concentrations of both reactants & products are present

(4) None of these

16. By the Equilibrium constant we can–

(1) predict the extent of a reaction on the basis of its magnitude

(2) predict the direction of the reaction

(3) Calculate equilibrium concentrations

(4) All of these

17. The liquid which has a higher vapour pressure is more volatile and has a–

(1) Higher boiling point

(2) Lower boiling point

(3) Higher force of attraction molecules

(4) None of these

18. Boiling point of the liquid depends on the–

(1) atmospheric pressure (2) amount of liquid

(3) Equilibrium constant (4) None of these

19. Temperature changes affect–

(1) The Equilibrium constant

(2) rates of reactions

(3) Equilibrium System

(4) All of these

20. Naphthalene, a white solid used to make moth balls has a vapour pressure of 0.10 mm Hg at 27°C. Hence K_p and K_c for the equilibrium are:

$C_{10}H_8(s) \rightleftharpoons C_{10}H_8(g)$

(1) 0.10 atm, 0.10 atm

(2) 0.10 atm, 4.1×10^{-3} atm

(3) 1.32×10^{-4} atm, 5.34×10^{-6} atm

(4) 5.36×10^{-6} atm, 1.32×10^{-4} atm

21. For the following equilibrium

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

K_p is found to be equal to K_c. This is attained when–

(1) T = 1 K (2) T = 12.18 K

(3) T = 27.3 K (4) T = 273 K

22. At a given temperature equilibrium is attained when 50% of each reactant is converted into the products

$A(g) + B(g) \rightleftharpoons C(g) + D(g)$

If amount of B(g) is doubled, percentage of B converted into products will be–

(1) 100% (2) 50%

(3) 66.67% (4) 33.33%

23. The position of the equilibrium for a system when $K_c = 4.6 \times 10^{-15}$ can be described as being favoured to........., the concentration of products is relatively.......

(1) The right, larger (2) The left, small

(3) the left, larger (4) the right, small

24. In the following equilibrium

I : $A + 2B \rightleftharpoons C$; $K_{eq} = K_1$

II : $C + D \rightleftharpoons 3A$; $K_{eq} = K_2$

III : $6B + D \rightleftharpoons 2C$; $K_{eq} = K_3$

hence, reaction between K_1, K_2 and K_3 is :

(1) $3K_1 K_2 = K_3$ (2) $K_1^3 K_2^2 = K_3$

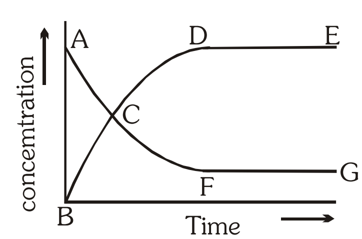
(3) $3K_1 K_2^2 = K_3$ (4) $K_1^3 K_2 = K_3$

25. For the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, $K_c = 4$. This reversible reaction is studied graphically as shown in the given figure. Select the correct statements out of I, II and III.

I : Reaction quotient has maximum value at point A

II : Reaction proceeds left to right at a point when $[N_2O_4] = [NO_2] = 0.1$ M

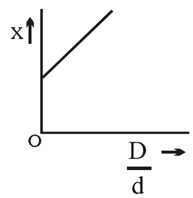
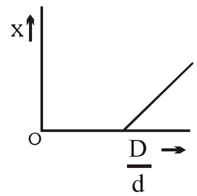
III: $K_c = Q$ when point D or F is reached.

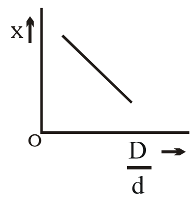
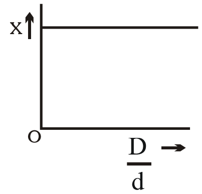


(1) I, II (2) II, III

(3) I, III (4) I, II, III

26. In the dissociation of N_2O_4 into NO_2. x varies with $\frac{D}{d}$ according to– (x = Degree of dissociation)

(1) 
(2) 

(3) 
(4) 

27. Consider following equilibrium in gaseous phase

I : $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

II : $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

III : $2CH_3COOH(g) \rightleftharpoons (CH_3COOH)_2(g)$

observed vapour density is larger than theoretical value in case of

(1) I, II (2) I, III
(3) II, III (4) I, II, III

28. The equilibrium constants for the reaction

H_3BO_3 + glycerine $\rightleftharpoons$ (H_3BO_3. glycerine) is 0.90.

Glycerine present per litre of 0.1 M H_3BO_3 to convert 60% of H_3BO_3 into (H_3BO_3. glycerine) is–

(1) 0.167 M (2) 1.67 M
(3) 0.0167 (4) 10.67 M

29. Increase in the pressure for the following equilibrium

$H_2O(\ell) \rightleftharpoons H_2O(g)$ result in the

I : Formation of more $H_2O(\ell)$
II : Formation of more $H_2O(g)$
III : Increase in boiling point of $H_2O(\ell)$
IV : Decrease in boiling point of $H_2O(\ell)$

Hence, correct statements are–

(1) I, II (2) II, III (3) I, III (4) I, IV

30. Formation of acetone is favoured for the following equilibrium if

$(CH_3)_2CHOH(g) \rightleftharpoons (CH_3)_2CO(g) + H_2(g)$, $\Delta H = +57.3$ KJ

(1) The temperature is increased
(2) The volume is increased at constant temperature
(3) $H_2(g)$ is withdrawn
(4) Under all the conditions given above

31. In each of the following total pressure at equilibrium is assumed to be equal and is 1 atm with equilibrium constant K_p given

I : $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, K_1

II : $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, K_2

III : $NH_2CO_2NH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$, K_3

Increasing order of K_p is –

(1) $K_1 = K_2 = K_3$ (2) $K_1 < K_2 < K_3$
(3) $K_3 < K_2 < K_1$ (4) None of these

32. ΔH for reaction does not affected by Catalyst because–

(1) It lowers the activation energy for the forward reaction.
(2) It lowers the activation energy for the forward and backward reactions by exactly the same amount.
(3) It decreases the rate of forward reaction.
(4) It does not change the rate of reaction.

33. The equilibrium constant for the hydrolysis of sugar is 2×10^{13} at 300K. The $\Delta G°$ of the reacttion is –

(1) -7.6×10^4 J.mol^{-1} (2) -15.8×10^4 J.mol^{-1}
(3) 7 6 10^4 J l^{-1} (4) 15 8 10^4 J l^{-1}